



Q.1 (C)

$$1 \leq l \leq m \leq n \leq 20$$

Cases

Number of triplets

Case-I: $l = m = n \rightarrow {}^{20}C_1$

Case-II: $l = m < n \rightarrow {}^{20}C_2$

Case-III: $l < m = n \rightarrow {}^{20}C_2$

Case-IV: $l < m < n \rightarrow {}^{20}C_3$

$$\therefore \text{Total number of triplets} = {}^{20}C_1 + 2 \cdot {}^{20}C_2 + {}^{20}C_3 = {}^{21}C_2 + {}^{21}C_3 = {}^{22}C_3 \equiv {}^pC_q$$

$$\therefore |p + q|_{\text{least}} = 25. \text{ Ans.}$$

Q.2 (B)

$$T_{r+1} \text{ of } (1 + x^3)^{15} = {}^{15}C_r x^{3r}$$

$$T_{m+1} \text{ of } (1 + x^2)^{10} = {}^{10}C_m x^{2m}$$

$$T_{r+1} \text{ of } (1 + x^3)^{15} (1 + x^2)^{10} = {}^{15}C_r {}^{10}C_m x^{3r+2m}$$

x^5 will form only if $r = 1, m = 1$

$$\therefore {}^{15}C_1 {}^{10}C_1 = 150 \text{ Ans.}$$

Q.3 (D)

$$T_{r+1} = {}^{30}C_r 2^{\frac{30-r}{2}} 3^{\frac{r}{3}} = {}^{30}C_r 2^{\frac{15-r}{2}} 3^{\frac{r}{3}}$$

$$\therefore \frac{r}{2} \text{ and } \frac{r}{3} \text{ should be integer}$$

$$r = 0, 6, 12, 17, 24, 30$$

$\therefore$ 6 terms are rational and 25 are irrational.

Q.4 (A)

$$\frac{31}{42} = \left(\frac{n}{2n+1} \right) \cdot 1 + \frac{(n+1)}{(2n+1)} \cdot \frac{1}{2}$$

$$\Rightarrow (31) \cdot 2(2n+1) = 42(2n+n+1)$$

$$\Rightarrow 31(2n+1) = 21(3n+1)$$

$$\Rightarrow 62n + 31 = 63n + 21$$

$$\Rightarrow n = 10 \text{ Ans.}$$

Q.5 (A)

$$f(x) = \frac{4(\sqrt{x+1} - \sqrt{3-x})}{x-1} = \frac{8(\sqrt{x+1} - \sqrt{3-x})}{(x+1) - (3-x)} = \frac{8}{\sqrt{x+1} + \sqrt{3-x}}$$

Domain of the function is $[-1, 3] - \{1\}$

$$\therefore \text{Range of the function is } [2\sqrt{2}, 4]$$



Hence, required probability is $\frac{2}{10} = \frac{1}{5}$ **Ans.**

Q.6 (C)

Q.7 (C)

Q.8 (A)

$$f(x) = \left| \sin\left((2r_1 - 1)\frac{\pi}{6}\right)x \right| + \left| \cos\left(\frac{\pi r_2}{6}\right)x \right|$$

$$f(1) = \left| \sin(2r_1 - 1)\frac{\pi}{6} \right| + \left| \cos\left(\frac{r_2 \pi}{6}\right) \right|$$

Possible value of $(2r_1 - 1)\frac{\pi}{6} = \frac{\pi}{6}, \frac{3\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{9\pi}{6}, \frac{11\pi}{6}$

Possible values of $r_2 \frac{\pi}{6} = \frac{\pi}{6}, \frac{2\pi}{6}, \frac{3\pi}{6}, \frac{4\pi}{6}, \frac{5\pi}{6}, \frac{6\pi}{6}$

(i) For $f(1)$ to be an integer

$$\underbrace{(2r_1 - 1)\frac{\pi}{6}}_{\frac{3\pi}{6}, \frac{9\pi}{6}}$$

$$\underbrace{r_2 \frac{\pi}{6}}_{\frac{3\pi}{6}, \frac{6\pi}{6}}$$

$$\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\frac{2\pi}{6}, \frac{4\pi}{6}$$

$$\therefore \text{Required probability} = \frac{2}{6} \times \frac{2}{6} + \frac{4}{6} \times \frac{2}{6} = \frac{12}{36} = \frac{1}{3}$$

$$(ii) f(2) = \left| \sin(2r_1 - 1)\frac{\pi}{3} \right| + \left| \cos\left(\frac{r_2 \pi}{3}\right) \right|$$

$$(2r_1 - 1)\frac{\pi}{3} = \frac{\pi}{3}, \frac{3\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{9\pi}{3}, \frac{11\pi}{3}$$

$$r_2 \frac{\pi}{3} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{6\pi}{3}$$

For $f(2)$ not to be an irrational number

$$(2r_1 - 1)\frac{\pi}{3} = \frac{3\pi}{3}, \frac{9\pi}{3} \quad \text{and} \quad \frac{r_2 \pi}{3} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{6\pi}{3}$$

$$P(\bar{E}) = \frac{2}{6} \times \frac{6}{6} = \frac{1}{3}$$

$$\therefore P(E) = 1 - \frac{1}{3} = \frac{2}{3} \text{ Ans.}$$

Note: E denotes event when $f(2)$ is an irrational number.



$$(iii) \quad f(x) = \left| \sin(2r_1 - 1) \frac{\pi}{6} x \right| + \left| \cos\left(\frac{r_2 \pi}{6}\right) x \right|$$

$$\downarrow \qquad \qquad \downarrow$$

$$T_1 = \frac{6}{2r_1 - 1}, \quad T_2 = \frac{6}{r_2}$$

$$\text{Possible values of } T_1: \frac{6}{1}, \frac{6}{3}, \frac{6}{5}, \frac{6}{7}, \frac{6}{9}, \frac{6}{11}$$

$$\text{Possible values of } T_2: \frac{6}{1}, \frac{6}{2}, \frac{6}{3}, \frac{6}{4}, \frac{6}{5}, \frac{6}{6}$$

$$\text{LCM of } T_1 \text{ and } T_2 = \frac{\text{LCM of } N^r}{\text{MCF of } D^r} = \frac{6}{3} = 2 \quad [\text{period of the function is either 1 or 2}]$$

For period to be 2

$$\underbrace{\frac{T_1}{6}}_{\frac{6}{3}} \qquad \underbrace{\frac{T_2}{6}}_{\frac{6}{3}, \frac{6}{6}}$$

$$\frac{6}{9} \qquad \frac{6}{3}, \frac{6}{6}$$

$$\therefore \text{ Required probability} = \frac{4}{36} = \frac{1}{9}$$

Q.9 (A)

Q.10 (A)

Q.11 (C)

A_r = Number of heads of A

B_r = Number of heads of B

$$= P(A_1 \cap B_0) \cup P(A_2 \cap B_0) \cup P(A_2 \cap B_1)$$

$$= P(A_1 \cap B_0) + P(A_2 \cap B_0) + P(A_2 \cap B_1)$$

$$= P(A_1) P(B_0) + P(A_2) P(B_0) + P(A_2) P(B_1)$$

$${}^2C_1 \left(\frac{1}{2}\right)^2 \times \frac{1}{2^3} + {}^2C_2 \left(\frac{1}{2}\right)^2 \times \frac{1}{2^3} + {}^2C_2 \left(\frac{1}{2}\right)^2 \times {}^3C_1 \left(\frac{1}{2^3}\right) = \frac{3}{16}$$

P(A and B ties a particular game)

$$= P(A_0 \cap B_0) + P(A_1 \cap B_1) + P(A_2 \cap B_2)$$

$$= \frac{1}{2^2} \times \frac{1}{2^3} + {}^2C_1 \frac{1}{2^2} \times {}^3C_1 \frac{1}{2^3} + {}^2C_2 \times {}^3C_2 \frac{1}{2^3} = \frac{5}{16}$$

P(A wins the game)

$$= \frac{3}{16} + \frac{5}{16} \times \frac{3}{16} + \left(\frac{5}{16}\right)^2 \frac{3}{16} + \dots \infty = \frac{3}{11}$$



Q.12 (ABCD)

For i^{th} player to reach in final keep a pool of 4 player including P_i and 3 player P_j where $j < i$.

$$\therefore P(A_4) = \frac{{}^3C_3}{{}^7C_3} \quad P(A_5) = \frac{{}^4C_3}{{}^7C_3} \quad P(A_6) = \frac{{}^5C_3}{{}^7C_3}.$$

Q.13 (ACD)

Each one should get $(4\lambda + 2)$ $\lambda \in w$ blankets

$$(4\lambda_1 + 2) + (4\lambda_2 + 2) + (4\lambda_3 + 2) + (4\lambda_4 + 2) = 32$$

$$4(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4) + 8 = 32$$

$$\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = \frac{24}{4} = 6$$

$$\text{Number of ways} = {}^{6+4-1}C_{4-1} = {}^9C_3$$

Q.14 (BC)

$$\boxed{\text{INDIA}} \text{ N I D O L} - \boxed{\text{INDIAN}} \text{ IDOL}$$

$$\therefore a = 6! - 5! \text{ and}$$

$$\boxed{\text{INDIAN}} \text{ IDOL} - \boxed{\text{INDIAN}} \boxed{\text{IDOL}}$$

$$b = 5! - 2!.$$

Aliter: INDIANIDOL

$$\boxed{\text{INDIA}} \text{ N I D O L} = 6! \quad \left. \begin{array}{l} \boxed{\text{INDIAN}} \text{ I D O L} = 5! \\ \boxed{\text{INDIAN}} \text{ I D O L} = 2! \end{array} \right\} \Rightarrow a = 6! - 5!$$

$$\Rightarrow b = 5! - 2!$$

$$a + b = 6! - 2, \quad a - b = 6! - 25! + 2! = 4 \cdot 5! + 2.$$

Q.15 (A) S; (B) P; (C) R; (D) P

(A) $f(x)$ = continuous but not differentiable

$$y = \{0, 1, 2\}$$

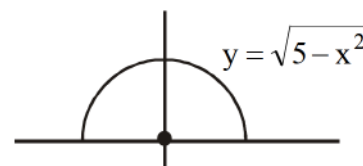
3 integers

(B) $g(x) = \text{sgn}(x^3 - 3x - 1)$

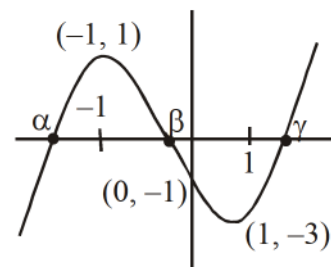
$g(x)$ is discontinuous at 3 points

$\therefore$ non differentiable at 3 points

Range of $g(x) = \{-1, 0, 1\}$ 3 integers



Graph of $x^3 - 3x - 1$



$$(C) \quad h(x) = \left[\sin^{-1} \left(\frac{2x}{1+x^2} \right) \right]$$

discontinuous and non differentiable at 5 points

$$\text{Range} = \{-2, -1, 0, 1\}$$

$$(D) \quad K(x) = (e^{|x|-1} - 1) |x^4 - x^2|$$

$$(e^{|x|-1} - 1) x^2 |x^2 - 1|$$

continuous everywhere & differentiable everywhere

Range contains ∞ integers

$$A : \{f, k\}$$

$$B = \{k\}$$

$$C = \{f, g, h, k\}$$

$$\bar{A} \text{ (selected function is discontinuous)} = \{g, h\}$$

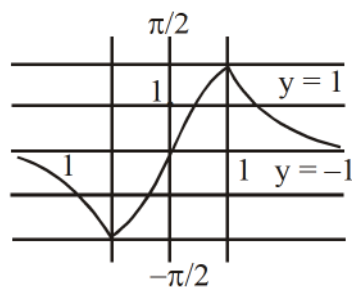
$$\bar{B} \text{ (selected function is non-derivable)} = \{f, g, h\}$$

$$P\left(\frac{\bar{B}}{A}\right) + P\left(\frac{A}{B}\right) = \frac{1}{2} + 1 = 3/2$$

$$P\left(\frac{B}{C}\right) + P\left(\frac{\bar{A}}{C}\right) = \frac{1}{4} + \frac{1}{2} = 3/4$$

$$P(\bar{B} \cap C) + P\left(\frac{A}{C}\right) = \frac{3}{4} + \frac{1}{2} = 5/4$$

$$P\left(\frac{C}{\bar{B}}\right) - P\left(\frac{(A \cap \bar{B})}{C}\right) = 1 - \frac{1}{4} = 3/4. \text{ Ans.}$$



Q.16 [5]

$$b_n = {}^nC_1 + {}^nC_2 \cdot a + {}^nC_3 \cdot a^2 + \dots + {}^nC_n \cdot a^{n-1}; \quad a = 4^{\frac{1}{401}} - 1 \Rightarrow 1 + a = 4^{\frac{1}{401}}$$

$$= \frac{1}{a} [{}^nC_1 \cdot a + {}^nC_2 \cdot a^2 + {}^nC_3 \cdot a^3 + \dots + {}^nC_n \cdot a^n]$$

$$= \frac{1}{a} [(1+a)^n - 1]$$

$$b_n = \frac{1}{a} [4^{n/401} - 1] \quad \dots (1)$$

$$b_{2006} = \frac{1}{a} \left[\left(4^{1/401} \right)^{2006} - 1 \right]; \quad b_{2005} = \frac{1}{a} \left[\left(4^{1/401} \right)^{2005} - 1 \right]$$

$$(b_{2006} - b_{2005}) = \frac{1}{a} \left[\left(4^{1/401}\right)^{2006} - \left(4^{1/401}\right)^{2005} \right] = \frac{1}{a} \left(4^{1/401}\right)^{2005} \underbrace{\left[4^{1/401} - 1\right]}_{=a}$$

$$= \left(4^{2005/401}\right) = 4^5 \equiv 4^k$$

$\therefore k = 5$. **Ans.**

Q.17 [1330]

$$\sum_{r=0}^{19} r(r+1) \frac{{}^{20}C_{r+1}}{{}^{20}C_r} = \sum_{r=0}^{19} r(r+1) \frac{(20-(r+1)+1)}{(r+1)} = \sum_{r=0}^{19} r \cdot (20-r) = \frac{20 \times 19 \times 20}{2} = 380. \text{ **Ans.**}$$

Q.18 [2]

m = number of ways in which girls are consecutive = $4! \cdot 5!$

n = number of ways in which exactly 3 boys are consecutive

$$= {}^4C_3 \times 4! \times {}^5C_2 \times 3! \times 2! = 4 \cdot 4! \cdot 5!$$

$$m + n = (1 + 4) \cdot 4! \cdot 5! = 5! \cdot 5! \quad \text{and} \quad p = 5! \cdot 5! \cdot 2!$$

$$\therefore \frac{p}{m+n} = 2 \quad \text{Ans.}$$

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